(Please write your Enrollment Number) Enrollment No. 19701182023

MID-TERM EXAMINATION
Program Name: B. Tech. Semester: 3
(October 2024)

Subject Code: BIT 201	Subject: Database Management System
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) Suppose that you are an Administrator (DBA) for a software application of Hospital Management System. What roles and responsibilities do you foresee for yourself for managing the complete database for this application?		CO1
	b) With reference to the File System and Database System, explain the logical data independence concept.		CO1
	c) Suppose you have to delete all the values from the DESIGNATION column of EMPLOYEE table. Which SQL Command will you use? Write the syntax also.		CO1
	d) Discuss 3 Layer Architecture of DBMS with the help of appropriate diagram		CO1
	UNIT I		CO Mapping
Q2	Attempt any two parts	(5*2=10)	
a)	In an organization several projects are undertaken. Each project can employ one or more employees. Each employee can work on only one project. Each project is undertaken on the request of a client. A client can request several projects. Each project have only one client. A project can use a number of items and an item may be used by several projects. Draw an ER diagram by clearly mentioning all assumptions required for designing the ER Diagram.		CO1
b)	Design an ER Diagram of Hospital Management System by considering the following entity sets: DOCTOR, PATIENT, NURSE, MEDICINE, BED, WARD Clearly mention all assumptions considered for designing the ER Diagram.		CO1
c)	Explain the Weak Entity Set and Specialization concept with the help of suitable example.		CO1
	UNIT II		CO Mapping
Q3	Attempt any two parts	(5*2=10)	
a)	(I) Differentiate between Inner Join and Outer Join. (II) Consider the following Relations: EMP (empid, ename, salary, designation, date_join, deptid),	1+(1*4)=5	CO2

	DEPARTMENT(deptid, dname, location, hod), PROJECT (pr_id, pname, pdetails, start_date, completion_date, empid) Write Relational Algebra Expression for the following queries: i) Display all those employee names and their designations who are Clerks and have joined before 1st January 2020. ii) Display all the project names managed by Mr Ramesh. iii) Display all the projects which are running in CSE Department. iv) Display all employees who are not holding any project.		
b)	(I) Differentiate between COMMIT and ROLLBACK. (II) Consider the tables of Q. 3 (a) II and write the SQL statements for the following queries: i) Display all those employee names and their designations who are Clerks and have joined before 1st January 2020. ii) Display all the project names managed by Mr Ramesh. iii) Display the number of employees working in IT Department. iv) Display only those department names where exactly 5 employees are working.	1+(1*4)=5	CO2
c)	(I) Explain Super Key and Candidate key concept. (II) Discuss Referential Integrity concept with the help of a suitable example.	2.5*2=5	CO2